

Research about two Ideas

1) Detailed Comparison:

Criteria	Smart Classroom Energy Management	Smart Resume Analyzer
Type of Project	IoT + Web-Based	Pure Software (Web + AI)
Main Focus	Energy saving & automation	Placement & resume improvement
Technology Stack	Sensors + Microcontroller + Backend + DB	Frontend + Node.js + Python + DB
Hardware Required	Yes (ESP32, PIR, Relay etc.)	No
Cost	₹800 – ₹2,500 (hardware cost)	Almost zero (only software)
Technical Depth	Moderate (if basic logic used)	High (ATS logic + NLP + role matching)
Innovation Level	Good (with real IoT)	Very Good (AI-based system)
Implementation Difficulty	Medium (hardware + wiring + debugging)	Medium (backend + Python integration)

Criteria	Smart Classroom Energy Management	Smart Resume Analyzer
Risk Factor	Hardware failure / wiring issues	Mostly coding errors (manageable)
Future Scope	Smart campus, smart city	Career tool, startup idea, AI enhancement
Team Comfort (Software Focused)	Less comfortable	More comfortable
SIH Perspective	Strong in Smart Campus domain	Strong in Skill & Employment domain

2) Which One is More Trending?

Currently trending in tech industry:

- AI-based systems
- Resume screening automation
- NLP applications
- Hiring automation tools
- SaaS-based platforms

Companies are actively building AI tools for:

- Hiring
- Resume screening
- Skill analysis
- Career guidance

So from a **market trend perspective**, Resume Analyzer is stronger.

3) Pros and Cons?

Idea 1: Smart Classroom Energy Management (With IoT)

Pros

- Practical real-world problem (energy saving)
- Hardware + software integration
- Good introduction to IoT systems
- Suitable for smart campus concept

Limitations

- Core logic remains simple (ON/OFF based on presence or timetable)
- Innovation mainly depends on hardware setup

More aligned with IoT/Embedded domain than core CSE

Good project, practical, but technically moderate from a CSE placement perspective.

Idea 2: Smart Resume Analyzer

Pros

- Directly related to placements and internships
- Solves a real student problem
- Uses AI / NLP (trending technologies)
- Shows full-stack development skills
- Demonstrates backend + database + API integration
- Scalable to ML-based enhancement
- Strong real-world industry relevance

Cons:

- Requires Node.js + Python integration

- Needs resume parsing logic
Slightly challenging, but much stronger technically.

4) SIH point of view?

i. Idea 1 (Smart Classroom – IoT)

Strong if:

- You use real sensors
- Add intelligent automation (not just ON/OFF logic)
- Include data analytics or optimization

Challenge for us (as Tech branch students):

- Hardware debugging
- Circuit issues
- Device reliability
- Integration complexity

At SIH level, many IoT energy solutions already exist.

So it must be *very advanced* to stand out.

ii. Idea 2 (Smart Resume Analyzer – AI Based)

Strong if:

- Resume parsing is accurate
- Scoring logic is meaningful
- Job-role matching is smart

Advantage for us (Tech branch students):

- Pure software implementation
- Uses backend + database + AI
- Easier to refine and improve
- Less physical risk

SIH panels usually appreciate:

- AI-driven impact solutions
- Youth employment-related problems

- Scalable digital platforms

Overall **summary**:

- From a **pure SIH competitiveness point of view**, **AI-based solutions currently have slightly higher chances**, especially when linked to employability and digital empowerment.
- From a **feasibility point of view (for CSE students)**, Idea 2 is safer, more controllable, and easier to enhance before submission.
- But if your team is genuinely interested in hardware + IoT and ready to build something advanced (not basic automation), Idea 1 can also work.

#Final Honest Advice

If our team:

- Has hardware knowledge
- Can manage sensors + wiring
- Has time for implementation

Then IoT-based Energy Management is powerful.

If our team:

- Is stronger in coding
- More comfortable with web + backend
- Wants AI exposure

Resume Analyzer is better.